

Data. Structural Validation Report

Task Information

Task ID: task1

Task Data Cleaning & Structural Validation

Internship: Data Analytics Internship – VirtualWorks Labs

Dataset Used: Dataset (Kaggle)

Tool Used: Microsoft Excel

1. Objective

The goal of this task was to clean and validate the dataset. This involved finding and fixing values checking for duplicate records and making sure the data is consistent. This process gets the dataset ready for analysis and keeps the data reliable.

2. Dataset Overview

The Titanic dataset has information about passengers. This includes Passenger ID, Name, Age, Gender, Ticket, Fare, Cabin and Embarked Port.

The dataset was checked for data quality issues before doing any analysis.

3. Data Cleaning Process

Missing Values

The dataset had missing values in some columns:

* Column: Age had 177 missing values. These were filled with the age.

- * Column: Cabin had 687 values. These were replaced with "Unknown".

- * Column: Embarked had 2 missing values. These were filled with the common value, which is "S".

Duplicate Records

The dataset was checked for duplicates using Excels Remove Duplicates feature.

Result: No duplicate records were found.

Structural Validation

The following checks were done:

- * PassengerId values are unique.

- * The Fare column has no values.

- * There are no missing values after cleaning.

- * Categorical values are consistent.

4. Results

After cleaning:

- * Missing values were handled.

- * No duplicates were found.

- * Data consistency and integrity were kept.

- * The cleaned dataset is ready for analysis and visualization.

5.conclusion

This task gave experience in finding and fixing common data quality issues with Microsoft Excel. Proper data cleaning and validation make analysis more reliable. Help produce accurate insights, from the dataset.